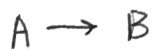


1a Single process catalyst and zeroth order



* Spherical and isothermal

$$C_A(r=R) = C_{As}$$

$$\frac{1}{r^2} \frac{d}{dr} \left(D_{AB} r^2 \frac{dC_A}{dr} \right) + k = 0$$

$$\int \frac{d}{dr} \left(D_{AB} r^2 \frac{dC_A}{dr} \right) = \int -kr^2$$

$$D_{AB} r^2 \frac{dC_A}{dr} = \frac{-kr^3}{3} + A$$

$$\int \frac{dC_A}{dr} = \int \frac{-kr}{3D_{AB}} + \frac{A}{D_{AB}r^2}$$

$$C_A(r) = B - \frac{A}{D_{AB}r^2} - \frac{kr^2}{6D_{AB}}$$

Boundary Condition

C_A is finite @ $r=0$

$C_A = C_{As}$ @ $r=R$

When C_A is finite @ $r=0 \rightarrow A=0$ ✓

$$C_{As} = \frac{-kR^2}{6D_{AB}} + B$$

$$B = \frac{kR^2}{6D_{AB}} + C_{As}$$

$$C_A(r) = \frac{k}{6D_{AB}} (r^2 - R^2) + C_{As}$$
 ✓

(i) (20/20)

(ii) (0/20)

(iii) (0/10)

(iv) (0/10)

bonus: (16/20)

Bouns

$$-D \frac{dC_A}{dr} = h_m (C_A - C_{AB})$$

$$C_A = -\frac{kr^2}{6D} - \frac{C_1}{Dr} + C_A$$

$$C_A = \frac{-kr^2}{6D} - \frac{h_m \frac{kR^2}{6D} + \frac{kR}{3}}{\frac{1}{R} + \frac{h_m}{DR}} \cdot \frac{1}{Dr}$$

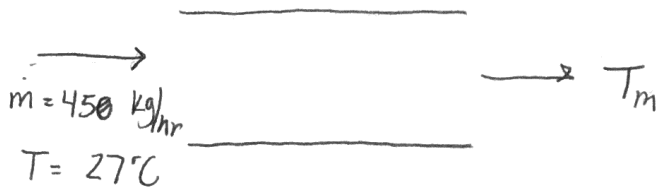
$$C_A = \frac{-kr^2}{6D} - \left(h_m \frac{kR^2}{6D} + \frac{kR}{3} \right) \left(\frac{1}{\frac{Dr}{R} + \frac{r h_m}{R}} \right)$$

$$\frac{dC_A}{dr} = -\frac{h_m}{D} (C_A - C_{AB})$$

$$-\frac{kR}{3D} + \frac{C_1}{DR^2} = \frac{-h_m}{D} (C_A - C_{AB})$$

almost!
x

2a



Energy Balance

$$dq_{\text{conv}} = m C_p dT_m$$

$$dq_{\text{conv}} = q' dx$$

$$q' dx = m C_p dT_m$$

$$dq_{\text{conv}} = q' dx$$

$$q' = m C_p \frac{dT_m}{dx}$$

$$\int_0^x q' dx = \int_{T_{m,i}}^{T_m} m C_p dT_m$$

$$T_m(x) = T_{m,i} + \frac{q' x^2}{2 m C_p}$$

2b

(a) $x = 30 \text{ m}$

$$T_m(30) = 27 + \frac{20(30)^2}{2 \left(\frac{450}{3600} \right) (4180)}$$

$$T_m(x) = 44.22^\circ\text{C}$$

